ODME USER GUIDE

Operation Domain Modeling Environment

A first-time user guide for running the program, understanding the workflow, and using the main modes without technical setup knowledge.

Who this guide is for: users who want to open ODME, create a project, build a model, validate it, and move into scenario work without reading source code or developer notes.

Product: Operation Domain Modeling Environment (ODME)

Purpose: Domain modeling, scenario modeling, validation, and supporting analysis tools

Recommended starting point: Domain Modelling mode

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1. What ODME Is

ODME is a desktop tool used to create and work with operation domain models and scenarios. Most users will spend the majority of their time in two modes:

- **Domain Modelling:** create the full structure of a model
- **Scenario Modelling:** turn the model into scenario-specific versions

The application also includes additional tools for behavior synchronization, scenario management, execution support, ODD work, and import from Cameo.

> **Important:** some of the supporting tools are still marked as work in progress. If you are new to ODME, begin with Domain Modelling and Scenario Modelling first.

2. How to Start the Program

Windows

1. Open the folder that contains the packaged ODME application.
2. Open the **ODME** folder.
3. Double-click **ODME.exe**.

macOS

1. Open the ODME package or disk image provided to you.
2. Open **ODME.app**.

Linux

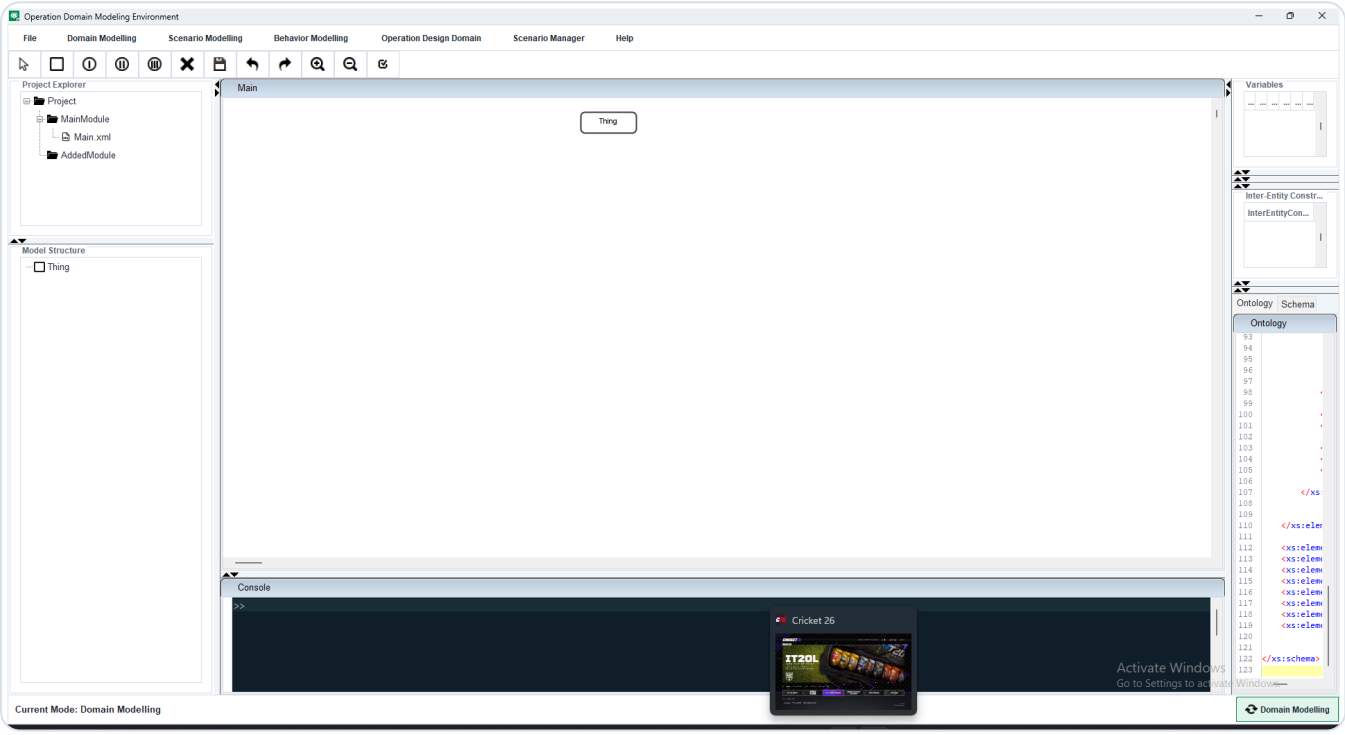
1. Open the ODME application folder.
2. Run the **ODME** launcher inside the **bin** folder.

Keep the whole application folder together. Do not move only the executable file. The program depends on other files next to it.

Best location: keep the ODME application in a folder where you have write permission, such as Documents, Desktop, or Downloads.

3. Main Window Layout

When the program opens, you will see these main areas:



Main ODME window in Domain Modelling mode. The left side contains the project and structure trees, the center contains the graph canvas, the right side contains the data panels, and the bottom area contains the console and viewer tabs.

Project Explorer
Shows the current project and related files or modules.

Model Structure
Shows the model as a tree. This stays synchronized with the graph.

Graph Canvas
The main drawing area where you build and edit the model visually.

Right-side Data Panels
Show variables, behaviors, distributions, and constraints for the selected node.

Console
Shows validation messages and operation feedback.

Viewer Tabs
Show XML, schema, or ontology output depending on the mode.

Status Bar
Shows the current mode and contains the mode-switch button.

4. Recommended Beginner Workflow

If this is your first time using ODME, follow this order:

1. **Step 1:** Start the program and stay in **Domain Modelling**.
2. **Step 2:** Create a new project.
3. **Step 3:** Build the basic domain structure on the graph canvas.
4. **Step 4:** Add variables, distributions, behaviors, and constraints as needed.
5. **Step 5:** Save the model and run validation.
6. **Step 6:** Switch to **Scenario Modelling**.
7. **Step 7:** Create or open a scenario.
8. **Step 8:** Prune the model and adjust scenario-specific values.
9. **Step 9:** Save or generate scenario outputs.

Simple rule: build the complete model first, then move to scenarios.

5. Domain Modelling Mode

Domain Modelling is the main authoring mode. It is where you create the full model structure before turning it into scenarios.

What you can do here

- Create a new project
- Open an existing project
- Add entities, aspects, specializations, and multi-aspects
- Edit the model tree and graph
- Add variables and distributions
- Add inter-entity and intra-entity constraints
- Sync or inspect behaviors
- Save and validate the model
- Export the model as a template
- Import from a template or from Cameo

Main toolbar actions

- Selector
- Add Entity
- Add Aspect
- Add Specialization
- Add Multi-Aspect
- Delete Node
- Save Graph
- Undo / Redo
- Zoom In / Zoom Out
- Validation

Typical menu actions

- **Domain Modelling > New Project**
- **Domain Modelling > Open**
- **Domain Modelling > Import Template**
- **Domain Modelling > Save as Template**
- **Domain Modelling > Import From Cameo**

Tip for first-time users: start by drawing the structure. Add variables and constraints after the structure is stable.

6. Scenario Modelling Mode

Scenario Modelling is used after the domain model is ready. It lets you work with scenario-specific versions of the model.

How to switch into it

Use the mode-switch button in the status bar at the bottom of the main window.

What changes in this mode

- Node creation actions are reduced
- The interface focuses more on scenario output and XML
- Scenario files and folders are created
- You can prune the model for specific scenarios

What you can do here

- Create and save scenarios
- Open scenarios from the scenario list
- Edit scenario-specific values
- Generate scenarios from CSV data
- Generate scenarios directly from the domain model

Good practice: if something feels wrong in Scenario Modelling, switch back and verify the base model in Domain Modelling first.

7. Other Modes and Supporting Tools

Area	What it is for	Advice for first-time users
Behavior Modelling	Sync or work with behavior-related data for entities.	Use after you are comfortable with domain and scenario workflows.
Operation Design Domain	Generate and manage ODD information based on the model.	Useful later in the workflow. Not required for first use.
Scenario Manager	Open scenario lists, execution helpers, feedback functions, and sample generation.	Use after you already have a working model or scenario.
Import From Cameo	Build a project from a set of required Cameo CSV files.	Only use if you already have the expected input files.

Area	What it is for	Advice for first-time users
Execution	Load XML or YAML, translate content, edit Python scripts, and run scripts.	Treat this as an advanced downstream tool.

Several supporting areas are still evolving. For reliable first use, stay focused on: **Domain Modelling** and **Scenario Modelling**.

8. Saving, Validation, and Export

Saving

Use the toolbar save action or the file menu to save the current project state.

Validation

Use the **Validation** toolbar button to check the generated model output. ODME will save, rebuild the XML and schema data, validate the result, and print details in the console.

Save as Template

Use **Domain Modelling** > **Save as Template** to export the model as XML for reuse.

Save as PNG

Use **File** > **Save as PNG** to export the visible graph as an image.

ODME now shows progress dialogs for longer validation and generation tasks. This means the window should stay responsive while these tasks run.

9. First Practice Session

If you want a short introduction session, do this:

1. Start ODME.
2. Create a new project.
3. Add a few nodes in Domain Modelling.
4. Select different nodes and watch the panels on the right change.
5. Save the model.
6. Run Validation.
7. Switch to Scenario Modelling.
8. Open the scenario list.

9. Create or generate one scenario.
10. Inspect the XML viewer and console output.

10. Troubleshooting

The program does not open

- Make sure you are opening the application from its full packaged folder.
- Do not move only the executable by itself.
- If you are running the jar directly, make sure Java is installed.

The program cannot save files

- Move the application to a writable folder.
- Avoid system folders such as Program Files unless you know you have write permission.

Validation or generation takes time

- This can be normal for larger models.
- Wait for the progress dialog to finish.
- Do not close the program while a long operation is running.

A supporting feature feels incomplete

- Return to the core workflow.
- Build and validate the model first.
- Use advanced tools only after the core model is stable.

11. Final Advice

The easiest way to learn ODME is to treat it as a two-stage process:

1. Create and validate the full structure in **Domain Modelling**.
2. Create and refine use-case-specific versions in **Scenario Modelling**.

If you stay with that sequence, the rest of the application will make more sense.